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PROJECT SCOPES &REQUIREMENTS

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AI-Supported Language and Speech Application

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T.C.

FATIH SULTAN MEHMET VAKIF UNIVERSITY

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GRADUATION PROJECT I REPORT

AI-SUPPORTED LANGUAGE AND SPEECH APPLICATION

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1) INTRODUCTION

Speech sound disorders (*articulation disorders*) are a significant problem in childhood, directly impacting an individual's communication skills, academic success, and psychosocial development. Traditional therapy methods, due to their confinement to the clinical setting, high costs, and transportation difficulties, present several sustainability limitations. The underlying scientific problem of this study is that when the therapy process is moved outside the clinic, particularly to the home environment, the controllability of standard treatment protocols and client motivation decrease.

The project developed to address this issue is based on the conceptual frameworks of "Gamification" and "Remote Health Monitoring." The methodology adopted a hybrid model that simulates client-therapist interaction on a digital platform. The technical infrastructure employed a three-layer data architecture (Configuration, Actors, Process) normalized on a PostgreSQL database management system to ensure relational integrity of data. The Unity game engine was used for the application interface and interaction mechanics, and the Java Spring Boot architecture was used for server-side operations. The study aims to expand the traditional "face-to-face therapy" paradigm with the "data-supported digital assistant" paradigm and accelerate the healing process by increasing the frequency of therapy repetition.

1.1) Problem Definition And Motivation

Language and speech skills acquired early in life have a decisive impact on an individual's cognitive development, academic success, and social integration. However, speech sound disorders (articulation disorders), which are common in childhood, can lead to permanent communication barriers if not intervened early. The relatively new nature of speech and language therapy in Turkey and the inadequate number of specialists relative to the population are significant factors limiting access to this service **[1]**. A review of current clinical practices and literature reveals the following fundamental scientific problems that limit the effectiveness of therapy processes:

Accessibility and Continuity Issues: Traditional speech therapy requires a physical presence in a clinic and can be costly. Due to geographical constraints or financial constraints, many children cannot access the therapy frequency they need. Studies in the literature indicate a high demand for, and a high level of satisfaction with, remotely accessible (teletherapy) models from clients and therapists **[2]**. However, appointment and cost barriers in current systems cause interruptions in the treatment process.

Loss of Motivation and Repetition Resistance: The success of articulation therapy depends on intensive repetition of correct sound production. However, repetition using traditional methods is known to decrease motivation in children. Research shows that "serious games" and gamification elements significantly increase children's attention and desire to participate in therapy **[3]**. However, most commercially available apps are either solely entertainment-focused or lack therapeutic objectives.

Lack of Data and Traceability: Tracking studies conducted outside the clinical setting is impossible with traditional methods. Clinical studies support the effectiveness of mobile-based systems in tracking the development of articulation skills **[4]**. The inability of therapists to monitor children's home performance with objective data makes it difficult to update the treatment plan based on data.

In light of, these scientific challenges, the primary motivation of the project is to extend the therapy process beyond the clinic walls and create a traceable, gamified rehabilitation model that is independent of time and space.

To implement this motivation, the project adopted the "*Gamification*" and "*Remote Health Monitoring*" paradigms, both proven effective in the literature, as conceptual frameworks. The methodology employed a hybrid structure that digitizes the interaction between therapist and client.

1.1.1) Project Main Purpose And Objectives

The main purpose of this project is to develop a digital speech and language therapy assistant that children aged 6 and older with articulation disorders can use both under the supervision of a therapist in a clinical setting and independently in a home environment. The system aims to overcome the limitations of traditional therapy methods, such as cost and accessibility, and to make the rehabilitation process sustainable, measurable, and accessible.

In line with this main goal, the project's sub-objectives are as follows:

Creating a Hybrid Therapy Model: The insufficient number of speech and language therapists in Türkiye relative to the population hinders access to therapy services **[1]**. This project aims to provide a hybrid structure that supplements clinic sessions with homework, enabling clients to achieve the repetition frequency they need and reducing therapists' workload.

Increasing Motivation with Gamification: It is well known that the "repetition" exercises that form the basis of articulation therapy can be tedious for children. The project aims to increase children's motivation and desire to participate in therapy by transforming syllable, word, and sentence-level exercises into gamified activities **[3]**.

Providing Data-Based Tracking and Feedback: To address the "data tracking" issue, a major shortcoming of home-based interventions, every game played by the child, their scores, and error rates will be recorded in a relational database. This will allow therapists to monitor their clients' progress with objective data and graphs rather than subjective statements **[4]**.

Ensuring Accessibility and Sustainability: The project aims to make the therapy process independent of the physical clinic environment through its mobile and web-based architecture. This will enable access to therapy from any location with internet access, overcoming geographical barriers and maintaining treatment continuity. Furthermore, the subscription models offered aim to create a more cost-effective and sustainable solution for both families and clinics compared to traditional methods.

1.1.2) Why Is This Project Needed?

Speech sound disorders are one of the most common communication disorders in childhood and, if not addressed early, carry the risk of negatively impacting a child's literacy skills, academic success, and psychosocial development **[5]**. When examining the current healthcare system and rehabilitation processes, the main reasons that necessitate this project can be categorized under four main headings:

Clinical Workload and Accessibility Constraints: Globally, and in Turkey specifically, the number of clients per speech therapist is quite high. According to data from the American Speech-Language-Hearing Association (ASHA), the caseloads of school-based therapists are increasing, leading to a decrease in the individual therapy time allocated to each child **[6]**. In Türkiye, the insufficient number of specialists leads to extended appointment times and inequalities in access to treatment **[1]**. The project offers a digital solution to this capacity problem by enabling therapists to provide services to their clients outside of the clinic.

Motivational Inadequacy of Traditional Methods: The success of articulation therapy depends on accurately repeating the targeted sound thousands of times (drill-play). However, static repetitions, such as those performed with paper materials or in front of a mirror, can quickly lead to a loss of attention span and resistance to therapy. Research shows that interventions using gamification techniques significantly increase child participation and the number of repetitions compared to traditional methods **[7]**. Most applications currently on the market either lack a therapeutic basis or lack Turkish language support and cultural adaptation; this project aims to fill this gap.

Data Blindness in Home-Based Practices: Therapists cannot objectively monitor how the child is doing at home after the clinical session, whether they completed their homework, or which sounds they made errors on. Relying solely on parental reports prevents data-driven updating of the treatment plan. Remote monitoring is a critical need for objective data collection **[4]**. The database architecture in our project eliminates this "data blindness" and makes the process transparent.

Economic Sustainability: Long-term therapy processes pose a significant financial burden for families. Both transportation costs and session fees can lead to treatment abandonment. This project, which offers a hybrid model, contributes to the economic sustainability of treatment by optimizing the frequency of clinic visits and providing a cost-effective digital alternative.

1.1.3 Project Users And Benefits

The developed "*Artificial Intelligence-Assisted Speech and Language Application*" aims to transform the therapeutic process from a solely patient-centered one to a multi-stakeholder ecosystem encompassing the child, therapist, and family. The user groups and key benefits of this project, which supports the "*Family-Centered Intervention*" approach in literature with digital tools, are detailed below:

1. Children (Players):

Motivation and Participation: The boredom of drill-based exercises in traditional therapy can create resistance to therapy in children. The project increases children's intrinsic motivation and reduces anxiety by presenting these exercises with gamification dynamics (points, levels, rewards) **[8]**.

Personalized Experience: Each child's developmental pace and pathology are different. The application offers a "flow" experience tailored to the child's current ability (neither too difficult nor too easy) thanks to the dynamic difficulty levels in the database.

2. Speech and Language Therapists (Specialists):

Data-Based Decision Making: Therapists struggle to monitor their clients' performance outside of the clinic. The system analyzes the child's error rates, practice times, and articulation success, providing the therapist with an objective "decision support mechanism." This facilitates the maintenance of evidence-based practices **[7]**.

Workload Management: Instead of paper-and-pencil-based assignment tracking, a digital assignment system reduces therapists' administrative workload, allowing them to dedicate more time to quality interaction during clinic sessions.

3. Families (Parents):

Economic and Logistical Access: Long-term therapy sessions pose a significant financial burden and transportation challenges for families. The hybrid model supports access to quality healthcare for every family, regardless of socioeconomic status, by optimizing clinic visit frequency and providing affordable digital access **[9]**.

Transparent Progress Tracking: Parents can track their child's progress in real-time through charts, becoming active supporters rather than passive observers of the therapy process.

1.2 Project Scope

This project encompasses an end-to-end software system consisting of a mobile-based client, server (backend), and relational database layers, developed to support the language development of children aged 6 and older with speech sound disorders (articulation disorders). Beyond being a mere game application, the project is a data-driven, hybrid (home-clinic) management panel that digitizes the clinical process between therapist and client, in accordance with the principles of "Digital Therapeutics." The system aims to eliminate inequalities in access to therapy materials and make repeat sessions sustainable through gamification.

1.2.1) Functional Components Of The System

The functional scope of the project includes the following modules, defined in the database architecture for three primary actors (Player, Therapist, Parent) and their authority over the system:

1. Game Module (Client Side): Equipped with visual and audio elements to engage children, this module combines therapeutic objectives with game mechanics.

Dynamic Content Management: As defined in the game_types table in the database, exercises are presented in three different linguistic hierarchies: Syllable, Word, and Sentence.

Adaptive Difficulty: Thanks to the algorithmic structure managed through the difficulty_levels table, the difficulty level of the games (Levels 1-3) can be gradually increased according to the child's performance.

Interaction Mechanics: Mechanics that simultaneously train motor and language skills, such as drag-and-drop, matching, and voice response, are used.

Feedback System: In-game reward mechanisms (stars, points) and instant visual/auditory feedback are provided to maintain the child's motivation.

2. Therapist Management Panel: This panel, developed for professionals to remotely monitor their clients' progress, is the system's administrative center.

Client Management: Therapists can add clients to the system and manage their profiles through the therapist_clients table.

Task Assignment: The therapist can specify which letter, game type, and asset set to work on, tailored to each child's needs, and record these in the tasks table.

Performance Analysis: Data from client-completed game sessions is analyzed, visualizing correct/incorrect response rates, completion times, and progress graphs.

Feedback: The therapist provides text-based feedback on completed tasks, enabling asynchronous communication with the family.

3. Parent Monitoring Interface: To ensure families' active participation in the process, an interface is provided where they can graphically track their child's daily and weekly work time and view progress reports sent by the therapist.

1.2.2) Game Scenarios And Content Details

The software modules developed within the project are based on four main game scenarios that target different cognitive and linguistic skills of children. These games are digitalized versions of therapeutic approaches in the literature:

Doll Dress-Up Game: This game aims to match words containing the target sound (e.g., "Hat," "Sweater") with visual objects. It is known in the literature that games based on "Daily Living Activities" support functional communication while developing children's vocabulary [10].

Greengrocer Game: This game targets the social use of language (pragmatics). With the "Customer-Seller" role-play scenario, the child is expected to understand auditory instructions (e.g., "I want an apple") and respond with correct pronunciation. Role-playing activities have been proven to increase children's communicative confidence [11].

Driving Game: This game transforms the "boring repetition"—the most challenging part of therapy—into a fun adventure. In the endless running game, the child is asked to repeat target sentences (e.g., "Rana bought lipstick") as the vehicle passes certain stops. The effect of game speed on motivation is supported by literature [12].

The "Taking Professions to Their Institutions" Game: This game aims to develop children's vocabulary through "semantic categories" (e.g., Doctor-Hospital matching). Semantic association and naming skills are simultaneously practiced with drag-and-drop mechanics [13].

1.2.3) Technical Scope And Architecture

The project is built on a multi-tiered (n-tier) architecture, in accordance with modern software engineering principles, to meet sustainability, scalability, and security requirements:

Mobile Application (Frontend): The game engine was developed using Unity. Designed for platform independence (Android/iOS), the interface is based on child-friendly user experience (UX) principles (large buttons, clear colors).

Backend: Business logic, authorization, and data processing processes were developed using the Java Spring Boot framework. Communication between the client and server is provided in JSON format via a RESTful API architecture.

Database: The PostgreSQL database management system was used for data consistency (ACID principles) and relational integrity. Designed in accordance with normalization rules, the schema stores Game Definitions (Configuration), Actors (Users), and Process (Interaction) data in three separate logical layers.

Artificial Intelligence Integration: While the current version of the system works with rule-based algorithms, the architecture is designed with flexibility to allow for automatic scoring of pronunciation accuracy by processing audio data collected in future phases (Future Work).

1.2.4) Exclusions And Limitations

To maintain the project's focus on "therapeutic rehabilitation" and ensure completion within the academic calendar, the following situations were excluded from the project scope:

Diagnosis and Prognosis: The application is not a medical diagnostic tool. It is an educational material that supports the rehabilitation process only for disorders diagnosed by specialist speech and language therapists.

Other Speech Disorders: The system currently focuses solely on "Articulation and Phonological Disorders." Other pathologies such as stuttering (fluency disorder), aphasia, or dysphonia are excluded.

Live Video Conferencing: Communication between the therapist and the client is conducted asynchronously (without synchronization) through data and report sharing. The application does not offer a synchronous live video conferencing feature similar to Skype/Zoom.

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2) RELATED WORK / LITERATURE SURVEY

This report examines the effectiveness of gamified mobile applications integrated with high-accuracy Automatic Speech Recognition (ASR) technology, a therapist management panel, and family participation features in the treatment of children with Speech Sound Disorders (SSD). Existing literature has been synthesized in light of epidemiological data, clinical trial results, and a deep analysis of the mobile application market. The analysis reveals the potential of digital interventions to bridge the "dosage gap" and increase the frequency of home practice. The report critiques the technological limitations (inadequate speech recognition) and procedural deficiencies (lack of therapist supervision) of existing market applications (e.g., Otsimo, Speech Blubs) and emphasizes the necessity of a hybrid model that combines high-accuracy ASR technology, therapist control, and family-centered features.

2.1) Introduction and Clinical Background

Speech Sound Disorders (SSD) are common communication disorders in childhood that place academic and social development at risk. While traditional therapy methods are effective, optimum treatment intensity (dosage) is often not achieved due to resource constraints.

2.1.1) Epidemiology and Prevalence of Speech Sound Disorders

Epidemiological data indicate that SSD has a significant prevalence in the pediatric population. According to US data, approximately 5% of children aged 3–17 experienced a speech disorder lasting a week or longer in the past 12 months **[14].** The prevalence of speech sound disorders (articulation or phonological disorders) in young children ranges from 8% to 9% **[14].** Persistent speech sound disorders, constituting approximately 8.8% of clinical caseloads, can evolve into permanent communication barriers if not treated early **[15]**.

Regarding gender and age distribution, boys (9.1%) are at higher risk compared to girls (5.2%), and prevalence reaches its highest level (10.8%) in the 3–6 age range. These data indicate that the preschool period is the most critical window for digital interventions. The most common phonological error patterns encountered in this period are detailed in **Table 1** below.

The most common phonological error patterns in children are:

Table 1: Common Phonological Error Patterns in Children [16]

Error Type	Prevalence Rate	Description
Fricative Simplification	82.5%	Replacing fricative sounds (f, v, s, z) with simpler sounds.
Cluster Simplification	49.0%	Simplifying consonant clusters (e.g., "train" -> "tain").
Gliding	41.3%	Transforming /r/ and /l/ sounds into sounds like /w/ or /y/.
Interdental Lisps	39.9%	Production of /s/ and /z/ sounds with the tongue between the teeth.

These data suggest that the speech analysis engine (API) of the application to be developed must be particularly sensitive to these sound groups (fricatives and liquids), as standard adult models may perceive these errors as "noise" or "accent" and correct them incorrectly **[17]**.

2.1.2) Limitations of Traditional Therapy and the "Dosage Gap"

Traditional speech therapy is typically a model based on tabletop activities and repetitive drills performed face-to-face with a Speech and Language Pathologist (SLP). Systematic reviews have proven this method to be highly effective, especially when combined with systematic interventions targeting entire sound classes in children with phonological process errors **[18]**. However, the effectiveness of therapy is directly related to "intervention intensity" or "dosage."

The fundamental problem defined in the literature is termed the "Dosage Gap." Motor learning principles and neuroplasticity studies argue that a high number of repetitions (e.g., 100+ trials per session) are necessary for acquiring and retaining a new speech skill **[19]**. However, due to constraints in public health systems, geographical barriers, and costs, children can typically access only one 30-45 minute session per week **[20]**.

Home Practice Compliance Issue: Therapists assign home exercise programs (HEPs) to families to bridge this intensity gap. Research shows that parent-implemented home programs can be as effective as traditional therapy if the dosage is equal to or greater than clinical therapy **[21]**. However, in practice, this compliance is not achieved. One study reported that 59% of families completed assigned home exercises less than 75% of the time **[22]**. Psychosocial barriers faced by parents, the child's lack of motivation, and the parent's uncertainty about the accuracy of the child's production (lack of expert feedback) are primary factors reducing the efficiency of home practice **[22]**.

At this point, a gamified mobile application can reduce the burden of "being the therapist" on the parent, provide the child with instant and accurate feedback, and thus raise the therapeutic dosage to clinically significant levels.

2.2) Methodological Foundations: Family-Centered Approach and Technology

2.2.1) Family-Centered Intervention (FCI)

Modern speech and language therapy is shifting towards the **Family-Centered Intervention** model. The American Speech-Language-Hearing Association (ASHA) defines the family as the "constant" in a child's development **[23]**. Systematic reviews show that when parents implement therapy techniques at home, children show progress that is greater than, or at least equivalent to, clinician-only therapy **[24]**. Digital tools support this approach by elevating parents from passive observers to active "co-therapists."

2.2.2) Gamification and Biofeedback

Gamification is critical for increasing motivation in motor learning processes that require repetition. According to Self-Determination Theory, games increase intrinsic motivation by satisfying children's needs for autonomy and competence **[25]**. In particular, visual biofeedback mechanisms (e.g., a character jumping according to the accuracy of the sound) help children concretize their own speech production and correct errors autonomously **[26]**.

2.3) Current Market Analysis and Evaluation of Applications

The global market reveals diversity in applications regarding their dependence on ASR (Automatic Speech Recognition) technology and their alignment with the family-centered approach. A comparative analysis of prominent applications regarding these criteria is provided in **Table 2** below.

Table 2: Comparative Analysis

Application Name	**Core Technology / Mechanism**	**Target Audience / Disorder**	**FCI (Family-Centered) Alignment**	**Clinical Evidence Level**	**Parental Role**
Otsimo	Voice Recognition Video Modeling	Autism, Delayed Speech, Apraxia	Moderate	Moderate (High User Satisfaction)	Data Monitor / Supporter
Speech Blubs	Video Modeling, Voice Activation	Early Language Development , Delayed Speech	Low - Moderate	Controversial (SLP Criticisms Exist)	Observer / Passive Participant
Splingo	Touch Interaction (No ASR)	Receptive Language, Following Directions	High (Together with Parent)	High (SLP Design)	Active Guide / Co-Player
Articulation Station	Digital Cards, Voice Recording (Human Evaluation)	Articulation Disorders, Phonology	Very High	High (Clinical Standard)	Active Co-Therapist / Evaluator

Stutter Stars	Customized ASR (Stuttering)	Stuttering	Moderate	Emerging (Clinical Research Based)	Supporter

2.3.1) Limitations of Current Applications

- **Speech Blubs and the Voice Activation Issue:** Most popular applications use "voice activation" (detecting the presence of sound) rather than phonetically based true ASR **[27]**. This leads to "false positives" where the application accepts it as "correct" and rewards the child even if they produce a wrong sound (e.g., "tcat" instead of "cat"). This is a serious risk for articulation therapy.

- **Otsimo and Articulation:** While possessing a large vocabulary pool, the therapist's ability to intervene instantly or assign specific "tasks" to the child's development is limited. Independent reviews emphasize that general-purpose ASR systems can be insufficient in detecting pathological errors in child speech **[28]**.

- **Articulation Station:** Although accepted as a "gold standard," it does not use ASR technology. Feedback depends entirely on the parent **[29]**. This increases the parent's burden and prevents the child from practicing autonomously.

2.3.2) Gap in the Turkish Market

While tele-practice usage is increasing in Turkey, **[30]**. the local market generally consists of simple vocabulary cards or digital versions of printed materials. There is a lack of an integrated platform that uses **high-accuracy ASR customized for Turkish** and allows the therapist to **assign tasks remotely and analyze results**.

2.4) Clinical Efficacy and Outcomes

2.4.1) Pronunciation Accuracy

The effectiveness of gamified therapy is supported by clinical studies. In the "Smart Speech" study conducted in Korea, Word-level Percentage of Consonants Correct (PCC) scores of children using the ASR-supported game for 4 weeks increased from **80.47** to **86.74** ($P=.003$), and these gains were maintained 8 weeks later **[31]**. Another study in Portugal found that tablet-based therapy was as effective as the traditional method **[32]**.

2.4.2) Speed and Efficiency

Applications have an indisputable advantage in increasing practice frequency (dosage) **[33]**. However, regarding "time to mastery," game design is crucial. A study comparing tablet-based intervention with traditional flashcards reported that the tablet group took longer to master target sounds (3 hours vs. 2 hours 15 min) **[34]**. This was due to in-game animations sometimes being distracting. This finding suggests that the proposed project's game design must establish an optimum balance between "entertainment" and "instruction."

2.4.3) Motivation and ASR Sensitivity

Gamification increases motivation, but this effect is strictly tied to the accuracy of ASR technology. False negative feedback (counting a child's correct pronunciation as wrong) can create frustration in children and damage trust in the therapy **[35]**.

2.5) Synthesis: Advantages of the Proposed Solution

Existing literature and market analysis confirm the strategic importance of the proposed project's positioning:

1. **Technological Superiority (True ASR):** Unlike applications like *Speech Blubs*, the proposed project uses advanced ASR (Speech Analysis API) technology to provide feedback not just because the child "made a sound," but because they "pronounced it correctly." This prevents incorrect learning and ensures therapeutic accuracy.

2. **Therapist Management Panel (Hybrid Model):** Unlike *Otsimo* and others, the ability of the therapist to **assign tasks** and analyze the child's pronunciation outputs transforms the application from merely a "game" into a clinical "tele-health tool." This feature fills the "Dosage Gap" while maintaining therapist supervision.

3. **Family-Centered Future Vision:** To address the lack of "Family-Centered Intervention" (FCI) in the literature, the **Multiplayer** mode in the project's future vision is critical. The parent's involvement in the game with the child will transform technology from a "digital babysitter" into a bridge that enhances intra-family interaction.

2.6) Conclusion

This research report confirms that the gamified mobile application to be developed for children with speech sound disorders, supported by ASR, therapist control, and family participation, is grounded in scientific foundations. The hypothesis is supported by strong evidence regarding pronunciation accuracy and motivation. The proposed system has the potential to offer an effective solution to the dosage problem in traditional therapy by addressing the technological (insufficient speech recognition) and procedural (lack of therapist connection) deficiencies of the existing market.

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3) METHODOLOGY AND SOLUTION TECHNIQUES

This section provides a detailed explanation of the general approach, multi-layered software architecture, relational data model, asset configuration (configuration) mechanism enabling the dynamic generation of games, and the objective pronunciation assessment process of the game-based speech and language therapy system developed for children with Speech Sound Disorder (KSB). The system adheres to current software engineering principles, consisting of educational game components developed with Unity and mobile application components developed with Flutter on the client side, and a Multi-Layered (N-Tier) Architecture comprising a.NET-based web service and a PostgreSQL relational database on the server side **[37]**.

3.1) Game-Based Speech and Language Therapy Approach

The fundamental approach adopted in the project is to move speech and language therapy beyond the confines of traditional paper-and-pencil or face-to-face clinical settings, conducting it through a gamified experience that captures children's interest. Children with KSB need intensive repetition focusing on specific target sounds at the syllable, word, and sentence levels; however, motivation tends to decline rapidly when repetitions become monotonous **[38][39]**.

In this context, the designed system offers a nine-step systematic game flow consisting of three different difficulty levels for syllable practice, three levels for word practice, and three levels for sentence practice, all selected based on the child's target sound(s) and aligned with clinical goals **[40]**. In addition to this individual, systematic flow, dual (co-play) modules designed to be played with a parent have also been incorporated to support family involvement (Family-Centered Intervention) in the therapy process and reinforce the social practice environment. These nine steps, corresponding to each target letter, are visually represented on a "path" screen, and the child marks the relevant tasks as active/completed upon finishing each step. Thus, the therapy process transforms from merely a series of exercises into a visible, rewardable game journey in the child's eyes **[41]**.

3.2) System Architecture

The system, supporting the principles of modularity, flexibility, and scalability, consists of three main layers: the client layer, the server layer, and the database layer **[37][42]**.

3.2.1) Client Layer: Unity Game and Flutter Mobile Application

The client layer includes two main components focusing on different user experience and functionality goals:

- **Unity-developed game module:** This is the main game environment where children directly interact, and the pet screen and nine-level game flow are executed. Visual and auditory interactions are provided within Unity using game templates for syllable/word/sentence exercises related to each target letter **[39]**. The Unity client retrieves JSON-based game configurations (level information, target sound, assets to be used, etc.) from the server and dynamically generates the scenes **[39]**.
- **Dual (Parent-Child) Practice Modules:** Extra games designed for parents to play with their children are available in the Unity interface, accessible independently of the individual scoring flow. The primary purpose of these modules is to involve the parent in the practice environment and increase social interaction; therefore, voice recording, Automatic Speech Recognition (ASR), and AI-based pronunciation assessment technology are not used. The pronunciation validation process relies directly on the parent's instantaneous feedback. This approach aims to increase treatment participation by supporting supervised home practice and family communication.
- **Flutter-developed mobile application:** This provides a separate interface designed for therapist and parent roles. Flutter's cross-platform development capability allows for rapid deployment to both iOS and Android platforms from a single codebase **[39]**.
- **Therapist Role:** Therapists can access the client list via the mobile application, view each child's past game sessions, pronunciation performance, and development curve. Additionally, the therapist can assign tasks for specific letters and game levels, managing the structural and procedural aspects of the therapy process **[40]**.
- **Parent Role:** Parents can monitor their child's game history, scores, and feedback recorded by the therapist; thus, the family can actively track the child's participation in the process **[39]**.

  Both clients connect to the.NET-based server via HTTP requests to retrieve necessary data and transmit updates via REST-like endpoints **[43]**.

3.2.2) Server Layer:.NET-Based Web Service

The server layer consists of a.NET technology-based web service that centrally executes the system's entire Business Logic. The fundamental responsibilities of this layer are:

- Executing user (player, therapist, parent) registration and Authentication processes.
- Managing therapist–player–parent relationships and coordinating data access.
- Providing game templates and asset configurations to clients.
- Recording the scores, pronunciation assessment results, and metadata of each game session to the database.
- Storing tasks assigned by the therapist, including target letter and level information, for task management.
- Communicating with the external pronunciation assessment service and processing the results.

  For this purpose, a layered architecture (Service-Repository pattern) separating the domain logic from controllers is preferred in the server layer. Service classes like GameService or AssignmentService access the PostgreSQL database via the Repository layer, which directly communicates with the database. This separation enhances code testability, sustainability, and independent development capability **[43]**.

3.2.3) Database Layer: PostgreSQL

PostgreSQL, an open-source relational database management system, is used for data management **[39]**. The database schema is designed with strict adherence to Third Normal Form (3NF) principles to ensure the reliability and consistency of sensitive clinical data **[44]**.

Role of 3NF in Clinical Data Integrity: The elimination of transitive dependencies (where a non-key attribute depends on another non-key attribute) mandated by 3NF minimizes data redundancy and prevents inconsistencies during data insertion, updating, and deletion operations **[44]**. This is crucial for healthcare informatics applications, which regard data as a strategic asset **[46]**.

The system schema is logically partitioned into Configuration, Actors, and Process layers, ensuring the relational integrity of critical clinical and game data **as detailed in Table 3 below**.

Table 3: Database Table Descriptions and Relational Rationale

Physical Table	Description and Relational Rationale
game_types, difficulty_levels, letters	Centralizes static, foundational high-level concepts (e.g., Syllable, Word, Level 1-3, Letters). All games connect to these tables using Foreign Keys (FK) to prevent data redundancy.
games	Defines the total nine fundamental game logics (templates) and their associated difficulty/type levels.
asset_sets	**Unique Constraint: UNIQUE(game_id, letter_id)**. Stores the JSON configuration (assets) used by a specific game template for a specific target letter, enabling **dynamic game generation.**
players	The system's single **Child (Player)** identity. All of the child's historical data is linked to this single ID, even if the therapist relationship changes.
therapists	Stores expert users authorized to assign tasks and provide feedback.
therapist_clients	Serves as a **Many-to-Many (M:N) relationship table**. Securely defines which player is a client of which therapist.
tasks	Records tasks assigned by the therapist. The asset_set_id field fixes the exact configuration at the time of assignment for **historical accuracy**.

game_sessions	The record of the child's **actual game playing event**. The **NULL** feature of task_id allows segregation between therapist-assigned tasks (populated) and free practice (NULL).
feedback	Records the therapist's comments and rating regarding a specific game_session (played session). Links directly to the session to maintain contextual integrity.

The data model is designed to facilitate detailed client-based monitoring by the therapist while also enabling the reuse of the same game templates for different letters and levels.

3.3) Game Design

3.3.1) Structure of Letters, Levels, and Game Templates

The system provides a systematic learning journey for each letter to align with the hybrid therapy approaches used by Speech-Language Pathologists (SLPs) **[40]**. This roadmap systematically represents therapy goals and difficulty levels:

- 3 syllable exercises (easy–medium–hard level)
- 3 word exercises (easy–medium–hard level)
- 3 sentence exercises (easy–medium–hard level)

Each step corresponds to a specific game template; for example, simple matching and repetition games for syllables, picture-word matching scenarios for words, and missing sentence completion or true-false sentence distinction for sentences **[39]**.

3.3.2) Asset Configuration Model and JSON-Based Game Generation

The fundamental methodological choice of the project is data-driven design. This involves defining a limited number of general-purpose game templates and reusing them dynamically with different asset configurations, rather than developing new games from scratch for every letter and level **[39]**.

Game templates function as "empty skeletons" with behavioral logic defined within Unity. Content specific to the letter, level, and target practice type (syllable/word/sentence) is stored in the asset and game_asset_config tables in the database **[39]**.

Dynamic Game Flow: The server selects the relevant configuration based on the "letter + level + game type" information received from the Unity client and transmits all content required by the game in JSON format to the client. This JSON structure includes rules such as the target letter and sound information, lists of syllables/words/sentences to be shown/targeted, paths to visual and audio files for each item, correct-incorrect answer pairings, and rules like how many repetitions will be performed or the achievement threshold for stars/badges **[39]**. The Unity client parses this JSON to generate the scene dynamically. This method:

- Allows the same game template to be reused repeatedly for different scenarios.
- Requires only updating the database by adding new asset and configuration records, eliminating the need to recompile the application to add new letters or exercises, which increases the speed of development and clinical adaptation.
- Reduces development costs while enabling the flexible enrichment of therapeutic content over time.

3.4) Pronunciation Assessment Component

The pronunciation assessment process is a critical component that enhances the educational value of the game and overcomes the limitations of traditional subjective assessment methods (e.g., VAS) **[47][48]**. When the child pronounces the relevant letter, syllable, word, or sentence, the system automatically analyzes this vocal performance to generate quantitative feedback **[49]**.

Methodologically, this system supports two different practice models: AI-supported quantitative scoring (individual games) and parent-guided subjective practice (dual games). Automatic quantitative analysis is only used in individual, isolated game modules, while pronunciation analysis is disabled in dual practice modules played with a parent.

For this purpose, the project utilizes an AI-supported pronunciation assessment service that evaluates short texts (posts) and corresponding voice recordings, returning data such as recording duration, number of words recognized, pronunciation score and reading speed classification for each word, and an overall pronunciation score (e.g., in the 0–90 range) **[39]**.

Conceptually, the process works as follows:

1. Reference Content Recording: The therapist or the system determines a short reference text for each exercise (e.g., the target word or a short sentence containing it). This text is recorded on the server side.
2. Acoustic Capture: When the child reaches the relevant task in the Unity game, the reference text is displayed on the screen, and the system records the child's voice via the microphone.
3. Service Call: The Unity client sends this recording to the server. The server transmits the recording file along with the reference text to the pronunciation assessment service.
4. AI Analysis: The service processes the recording using Automatic Speech Recognition (ASR) and pronunciation analysis algorithms **[39][50]**. This analysis is built upon acoustic feature extraction and deep learning-based acoustic models **[45]**. The service returns a structured result:
    - Pronunciation score for each word (e.g., score between 0–90),
    - Reading speed assessment per word (classifications like good/slow/fast/not available),
    - An overall pronunciation score in the 0–90 range and summary information about the recording.
    - The overall score is based on a special scoring algorithm developed by the service provider, rather than a simple average of the word scores.
5. Data Persistence: The server records the incoming results, linking them to the relevant game session and the child's record in the PostgreSQL database.
6. The Unity client can use these scores to provide instantaneous in-game feedback (successful/areas for improvement, etc.); the therapist and parent can access the summary and detailed views of the results via the Flutter application.

This approach ensures that pronunciation assessment moves beyond relying solely on subjective observation, achieving comparable quantitative metrics over time **[48]**.

3.5) Data Flow and Game Session Management

When each game is launched, a "game session" (session) record is opened by the server layer. This record forms the center of the database schema's hierarchical structure **[51]**:

- **Session Metadata:** Player identity, relevant letter, level and game type, session start time, and information on whether it is linked to a therapist-assigned task (task_id) are stored.
- **Granular Recording and Practice Model:** Attempts, points earned, and scores returned from each pronunciation assessment call during the game are associated with this session. Since voice recording and the GOP score are not recorded in the dual (parent-child) practice modules, these sessions are processed in the database by leaving the score field NULL or by using a simple completion/success tag provided by the parent. This ensures that the clinical validity of subjective scores does not compromise the objective scoring data, while still allowing the monitoring of parent involvement and practice intensity.
- **Final Metrics:** Upon completion of the session, summary metrics such as the final game score, overall pronunciation score, correct pronunciation rate per word/target, and session duration are calculated and recorded in the database **[39]**.

This session management creates the scientific database that enables the extraction of both single-session performance and time-based development graphs for the same child, categorized by letter, level, or game type **[39]**.

3.6) Security, Authentication, and Role Management

The system features three fundamental user roles (player, therapist, parent). Due to the necessity of managing sensitive health data (PHI), security and data integrity are fundamental architectural requirements.

- Centralized Identity Management: All user accounts are managed on the server side, and application access is provided via strong authentication mechanisms **[52]**. Passwords are stored using irreversible hashing algorithms (such as BCrypt) to prevent plain text storage **[52]**.
- Role-Based Access Control (RBAC): Authorization is implemented on the server side using Role-Based Access Control **[53]**:
- Therapists have the authority to view all their clients and assigned tasks.
- Parents can only view data related to their own children.
- This segregation is ensured by filtering the endpoints called by clients based on the user's role and associated children. This method ensures the protection of personal data and maintains transparency in the limits of authority within the therapeutic process.

3.7) Suitability and Sufficiency of the Method

The defined methodology and solution techniques align directly with the project's main goals, demonstrating a high degree of suitability and sufficiency **[54]**:

- **Clinical Suitability and Systematic Practice:** The pet-based nine-step game flow ensures that children with KSB can work systematically and progressively at the syllable, word, and sentence levels **[40]**. Furthermore, the dual practice modules support generalization to the social environment and parent involvement.
- **Flexibility and Sustainability:** The Unity-based game templates and JSON-based asset configuration structure allow for the flexible definition of numerous therapy scenarios, supporting rapid adaptation.
- **Data Reliability and Performance:** The.NET-based backend, PostgreSQL, and 3NF principles ensure that therapist, parent, and child data are managed securely and consistently **[44]**.
- **Objective Measurement Sufficiency:** The AI-supported pronunciation assessment component generates numerical indicators that enhance therapy effectiveness and provide objective feedback for both the therapist and the family **[48]**.

With this holistic approach, the project aims not only to develop a game application but also to establish a data-driven, traceable, and sustainable digital ecosystem that supports speech and language therapy.

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4) DEVELOPED APPLICATION AND FINDINGS

This section presents the implementation of the "Artificial Intelligence-Assisted Language and Speech Application," whose architecture and design details were described in previous sections. The system consists of two main interfaces: a "Mobile Game Interface," where end-users, children, interact, and a "Therapist Management Panel," where experts manage the process.

4.1) Mobile Game Interfaces (Client-Side)

The mobile app was developed with a simple and intuitive design that adheres to user experience (UX) principles, taking into account the motor skills and attention spans of children ages 6 and older. Built on the Unity game engine, the interfaces feature vibrant colors and large buttons to boost children's motivation.

4.1.1) Login and Main Menu

The login screen that greets the user upon application launch ensures a secure entry into the system. After logging in, the main map screen appears, where game types (Syllable, Word, Sentence) and levels can be selected. Games are presented in a hierarchical structure. Users can view locked tasks or free play modes assigned by their therapist through this menu.

4.1.2) In-Game Interactions

The "Drag-and-Drop" mechanic was adopted as the primary interaction method in the developed game modules (Baby Dressing, Uncle Greengrocer, etc.). The image below shows a section of the "Toy Matching" module.

Figure 1: Matching Toys game and drag-drop

In Figure 1, when the user drags the visual object to the correct target, the system checks the accuracy of the match with the algorithm running in the background and provides instant visual feedback (confetti, score increase). During this process, the game's completion time and success status are recorded instantly in the game_sessions table in the database.

4.1.3) Game Scenarios and Interfaces

Memory Card (Matching) Game: In Figure 2, this game improves children's visual memory and attention span while encouraging them to repeat words containing the target sound (e.g., "Cat," "Dog," and "Frog" for the sound "K" in the visual). Gameplay:

- Cards are placed face down on the screen (a 2x3 grid in the example).
- When the child touches a card, the card is turned face up, and the system vocalizes the name of the object on it (Auditory Input).
- The child is asked to find the matching card and repeat the resulting word (e.g., "Frog").
- When a correct match is made, the cards remain face up, and a visual/auditory reward is awarded. Technical Infrastructure: This game dynamically retrieves images for the currently studied letter (Letter ID) from the asset_sets table in the database. For example, if the child is studying the letter "S," the cards are loaded with images of "Clock" or "Snail" instead of "Cat."

Figure 2: Memory Card Game interface

Shadow Guessing (Silhouette) Game Purpose: To encourage children to recognize objects by their shapes using their visual perception skills and to correctly pronounce the corresponding word (e.g., "Cat"). This game stimulates curiosity and turns articulation into a fun puzzle.

Gameplay:

- Phase 1 (Exploration): Only the black shadow (silhouette) of an object containing the target sound appears on the screen. The child is asked to guess what this shape is.
- Phase 2 (Production): The child presses the microphone button and says the word they guessed (e.g., "Cat").

- Phase 3 (Reward): If the pronunciation is verified by the speech recognition system (or therapist approval), the black silhouette disappears and the actual, colored version of the object appears with applause/confetti effects.

Technical Infrastructure: In Figure 3, this game type defined in the games table, the system pulls two different images (one black mask, one colored original) from the asset_sets table. In the Unity interface, the transition between these images is triggered based on the user's voice input.

Figure 3: Initial silhouette view

4.2) Current Implementation Status and Progress

Following the initial architectural design, the system has successfully transitioned from the development environment to a live cloud infrastructure. The implementation details regarding the cloud deployment, asset management, and the finalized user interfaces are described below.

4.2.1) Cloud Infrastructure and Data Management

To ensure high availability and accessibility, the server-side components have been fully deployed to a cloud environment:

- **Backend and Database Hosting:** The Java Spring Boot backend service and the PostgreSQL database are now hosted on the **Render** cloud platform. This migration ensures that the API endpoints are accessible to mobile clients over the internet, enabling real-time data synchronization.

- **Asset Management (Cloudinary):** To optimize the mobile application's file size and allow for dynamic content updates, game assets (visuals and audio files) are stored in **Cloudinary**, a cloud-based media management service.
- **JSON Configuration Flow:** The asset references stored in Cloudinary are serialized into JSON format. These JSON configurations are hosted and served by the backend on **Render**. When a child starts a game, the client dynamically fetches the relevant JSON file to load the correct assets (e.g., images for specific letters) without requiring an app update.

4.2.2) Unity Game Modules and UI Enhancements The client-side application, developed using the Unity game engine, was designed with a focus on child-friendly UX principles, utilizing vibrant colors and intuitive navigation. The interface flow was structured to guide the child from secure login to specific therapeutic exercises.

Figure 4: User Login Interface

Login and Authentication To ensure data privacy and load specific user profiles, a secure login interface was implemented. As shown in **Figure 4**, the user is required to enter their credentials to access the system. This step is crucial for fetching the correct "Active Tasks" and historical data from the backend.

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Figure 5: Level Difficulty Selection Map

Navigation and Progression System Upon successful authentication, the user is presented with a hierarchical map system. First, the user selects the difficulty tier (Easy, Medium, Hard) appropriate for their therapy level, as illustrated in **Figure 5**.

Figure 6: Letter-Based Progression Pat

Following the difficulty selection, a gamified progression path appears. This interface, shown in **Figure 6**, visualizes the therapy curriculum as a journey, where each node represents a specific target letter (e.g., B, C, Ç, D). This visual structure provides the child with a sense of continuity and achievement.

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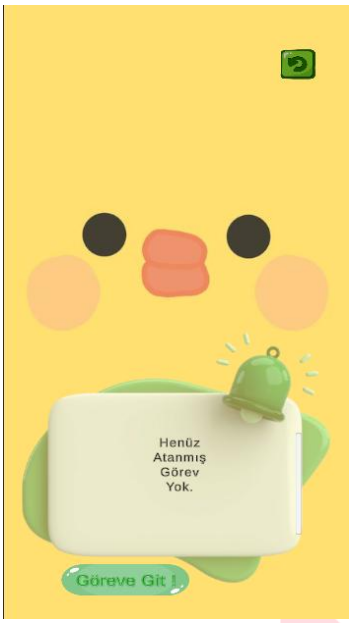

Figure 7: Task Notification Screen

Task Management A dedicated notification interface was developed to inform the user about therapist-assigned exercises. As seen in **Figure 7**, this screen provides clear feedback (e.g., "No Assigned Tasks Yet") or directs the user to specific active assignments, ensuring the child focuses on the correct therapeutic goals.

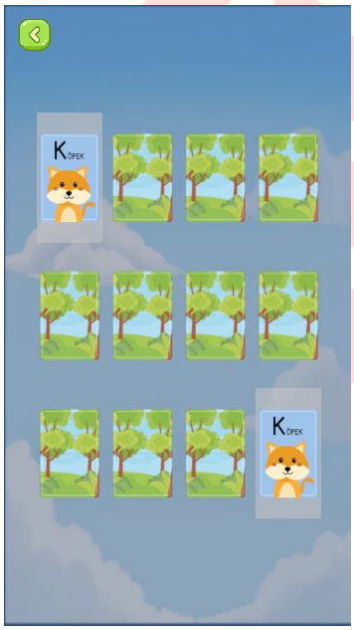

Figure 8: Memory Game Interface

Dynamic Gameplay: Memory Game The "Memory Game" module was implemented to function dynamically based on the selected letter. As demonstrated in **Figure 8**, the game board and card assets are not hard-coded; instead, they are populated at runtime using the JSON configuration fetched from the Render server. For instance, if the target letter is "K", the cards automatically display relevant images (e.g., "Köpek" - Dog) stored in Cloudinary.

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5)PROJECT MANAGEMENT

Table 4: Project Development Phases, Milestones and Verification

Phase/Milestone	Defining Objective	Estimated Duration (Sprint)	Verification Output
Phase 1: Core Architecture Setup	Stabilization of the core Unity -> Spring Boot -> PostgreSQL data flow and JWT security.	4 Weeks (2 Sprints)	API test of basic CRUD operations, 3NF Data Model Schema, JWT validation test .
Phase 2: ASR/GOP Integration	Successful integration of the third-party OKT (ASR) service with Unity and transmission of the GOP score with acceptable latency to the API.	6 Weeks (3 Sprints)	Latency test results (Targeted sub-300ms instantaneous feedback), First GOP score recorded to the database [53].
Phase 3: Clinical Module Development	Completion of the nine-level game template, with dynamic Asset Management (JSON configuration), for the first target letter.	8 Weeks (4 Sprints)	Completion of the first therapist-assigned task (Task) record with a successful session (Session) .
Phase 4: Beta Testing and Validation	Usability Tests with Children, system security audit, and final acceptance tests by clinical stakeholders.	4 Weeks (2 Sprints)	Drop-off Rate and Frustration Points report, SLP final acceptance test protocol [54].

6) RISK MANAGEMENT

Table 5: Project Risk Analysis and Mitigation Strategies

Risk Description	Category	Potential Impact (Clinical/Technical)	Mitigation Strategy
Service Latency	Technical/External Dependency	Child motivation drop, instantaneous feedback corruption (UX error).	Use of Streaming APIs, Connection optimization, Fault Isolation [52].
Scope Creep	Managerial	Exceeding project duration and cost.	Clear definition of Exclusions, Use of Asset Configuration Model (Limiting content scope) .
Data Integrity Violation (3NF)	Technical/Data	Reporting inaccuracy, incorrect tracking of clinical progress.	Foreign Key constraints, periodic data validation tests, transaction management [53].
Child User Usability Issue	Quality/UX	High abandonment rate (Drop-off Rate), reduction in learning effectiveness.	Early and frequent repeated Play-Along and Think-Aloud tests with tracking of non-verbal signals [54].

7) BROADER IMPACT AND SCIENTIFIC CONTRIBUTION

7.1)Transformative Impact on Clinical Methodology and Therapy Processes

The primary and most significant impact of the project is its provision of an evidence-based solution aimed at increasing therapy intensity (dosage), a critical factor in ensuring clinical efficacy.

7.1.1. Closing the Dosage Gap and Ensuring Treatment Continuity

In traditional articulation therapies, achieving the high-dose repetition count necessary for motor skill acquisition is challenging due to clinician availability, costs, and logistical barriers. This situation results in a "Dosage Gap" that reduces treatment efficacy **[55]**.

- **Increasing Intensity:** The developed mobile platform aims to close this gap by enabling high-intensity, self-managed practice unrestricted by clinician availability. Research conducted on neurological disorders like aphasia indicates that digital health technologies can facilitate high-intensity, self-managed practice, improving functional communication recovery potential **[55]**.
- **Treatment Adherence:** Monitoring a child's progress and therapist feedback via the mobile application allows parents to actively participate in the treatment process.

7.1.2. Objective Decision Support

The project shifts therapists' decision-making processes from subjective observations to objective data.

- **Quantitative Data Integration:** Through AI-supported pronunciation analysis, objective, quantitative data, such as scores and trial counts related to the child's home performance, is captured. This granular data allows the therapist to update the treatment plan in a data-driven manner.
- **Clinical Validation:** The high consistency of ASR/GOP scores with Speech-Language Pathologists' (SLP) evaluations on fundamental clinical metrics like the Percentage of Consonants Correct proves that the system's objective scores have sufficient scientific validity to guide clinical decision-making processes **[56]**.

7.2) Socio-Economic and Accessibility Contribution

The developed mobile platform has a significant broader impact by reducing geographical and economic disparities in access to healthcare services.

7.2.1) Overcoming Geographical Barriers

- **Access to Remote Areas:** Teletherapy (remote therapy) solutions improve access to critical services and overcome transportation challenges for patients living in remote areas or facing mobility barriers, especially where specialist numbers are low. This allows patients to access therapy sessions without needing to travel to a clinic.
- **Cost-Effectiveness:** Flutter's cross-platform architecture, by enabling the development of iOS and Android applications from a single codebase, reduces development time and costs. This, in turn, helps deliver such mobile health solutions faster and more cost-effectively for clinical use **[57]**. This contributes to the long-term economic sustainability of therapy by reducing overall costs.

7.2.2) Optimizing Workload and Focusing on Service Quality

- **Therapist Workload:** Mobile health solutions have the potential to reduce the administrative burden on clinical service providers (SLPs) by automating repetitive administrative tasks (homework tracking, scoring). This allows therapists to dedicate more time to high-quality, one-on-one interaction with their patients.

7.3) Scientific and Technological Contribution

The project adheres to the core architectural principles of Medical Intervention Software, which aims to deliver a clinically proven treatment, rather than merely an entertainment or educational application. Since this type of software directly aims to alleviate a condition (SSD), it is subject to strict legal and scientific standards regarding efficacy, quality, security, and patient privacy **[58]**.

7.3.1) Clinical Software Security Architecture and ASR Contribution

The scientific contribution of this system is concentrated in two main areas:

- **Security and Compliance:** A robust security architecture is employed to protect sensitive patient data. The integration of industrial security mechanisms like Role-Based Access Control (RBAC) and JSON Web Token (JWT) protects the confidentiality of Protected Health Information (PHI) **[59-61]**. This layered security structure demonstrates technical compliance with the stringent privacy and quality standards required by regulations like KVKK/GDPR , thereby enhancing the project's clinical acceptability and scientific reliability **[61][62]**.
- **Application of ASR Technology to Disordered Speech:** The project presents a significant academic case for applying Automatic Speech Recognition (ASR) technology, one of its most challenging areas of use, to disordered children's speech **[56]**. Despite existing difficulties in recognizing the speech of children with Speech Sound Disorder , the use of ASR/GOP scoring demonstrates the technology's potential for therapeutic evaluation. This work provides a fundamental reference point, paving the way for future research (refining the model with more diverse voice samples and clinical settings) **[56]**.

7.4) Future Work Areas and Expansion Potential

The current modular and flexible architecture is a foundation ready for future research and development, designed to extend the success achieved in treating Speech Sound Disorder to other clinical domains and to secure technological independence.

7.4.1) Strategic Expansion of Clinical Scope

The project has established a strong foundation for the target population with Speech Sound Disorder, a condition with a high prevalence rate (approximately 8% to 10% in the child population) **[63]**. The platform's flexible asset management and task assignment structure allow for systematic adaptation to treat other clinically related disorders.

- **Childhood Apraxia of Speech:** Childhood Apraxia of Speech, a neurodevelopmental subtype of Speech Sound Disorder, requires more intensive repetition and specific feedback protocols **[58]**. The current system, through its dynamic task assignment mechanism, is capable of digitally modeling these intensive protocols. This presents a potential to fill a specialized gap in the market for childhood speech disorders **[64]**.
- **Neurological Speech Disorders:** The system can be easily adapted for neurological speech disorders such as Aphasia and Dysarthria, which occur in adults following stroke and brain injury. High-intensity, self-managed mobile therapy applications have been proven successful in the literature for these patient groups **[65]**. The project's core pronunciation analysis (GOP) kernel has the potential for application to multiple correlated disorders, extending the market potential beyond the pediatric population.
- **Academic Dissemination and Validation:** The platform provides a reliable, quantitative data collection infrastructure for future Randomized Controlled Trials (RCT). This supports the project's goal of contributing to the scientific literature by producing clinical efficacy evidence, not just commercial value.

7.4.2) Internalization of Artificial Intelligence Models

The system's current dependency on an external Automatic Speech Recognition service creates operational cost, latency, and data sovereignty risks in the long term. The future roadmap aims to eliminate this dependency.

- **Performance and Cost Efficiency:** The long-term development of an in-house Automatic Speech Recognition engine will significantly enhance the system's cost-effectiveness and real-time performance by eliminating third-party API costs and reducing network latency **[66]**.
- **Staged Training Approach:** The development of the internal Automatic Speech Recognition model will involve a staged approach, starting with high-quality, clean speech data and progressively introducing samples with noise and disordered speech to increase the model's robustness **[67]**. This will maximize its ability to accurately diagnose pronunciation errors specific to Speech Sound Disorders.

- **Technological Independence:** Possessing its own Automatic Speech Recognition engine will serve as a critical leverage point for expanding the system into international markets (supporting pronunciation analysis in different languages) and gaining a competitive advantage.

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8) CONCLUSION

This project presents a comprehensive, data-driven, and clinically aligned digital solution developed for children with Speech Sound Disorders (SSD). By combining gamification, remote monitoring, therapist-guided task management, and AI-supported pronunciation assessment within a multi-layered software architecture, the system successfully addresses the fundamental limitations of traditional therapy models—namely the dosage gap, accessibility barriers, lack of home-practice data, and motivational challenges.

The developed platform demonstrates that a hybrid therapeutic model, in which clinical sessions are supplemented by structured, measurable home practice, can improve treatment continuity and increase children's active engagement. The integration of objective pronunciation scoring creates a significant clinical advantage by transforming subjective observations into quantifiable progress indicators. Through this structure, therapists gain the ability to make data-supported decisions, parents become active partners in the treatment process, and children engage with exercises in a motivating and developmentally appropriate game environment.

From a technical perspective, the modular architecture built on **Unity** (for the child interface) and **Flutter** (for the therapist/parent dashboard) ensures a seamless cross-platform experience. The backend infrastructure, powered by **Java Spring Boot** and hosted on **Render**, along with a normalized **PostgreSQL** database, provides the necessary scalability and real-time data synchronization. Furthermore, the integration of **Cloudinary** for media management and the **JSON-based** asset configuration model enables efficient, dynamic content expansion without requiring frequent application updates. The secure RBAC-based authentication system ensures compliance with clinical data privacy requirements. Together, these design choices establish the foundation for a robust, cloud-native clinical software ecosystem rather than a simple educational application.

In broader societal terms, the system has the potential to reduce inequalities in access to therapy services, particularly for families facing geographical, economic, or resource-related challenges. Its suitability for clinical extension—such as apraxia, phonological disorders, or even adult neurological rehabilitation—further increases its strategic value. Future work,

including the internalization of ASR models and the expansion of clinical modules, will strengthen the project's scientific contribution and technological independence.

Ultimately, this project demonstrates that digital therapeutics, when built on clinical principles and rigorous software methodology, can meaningfully enhance the effectiveness of speech and language therapy. By bridging the gap between clinic and home, combining motivation with measurable performance, and ensuring therapist oversight through data-driven insights, the system represents an innovative, scalable, and clinically impactful solution for the treatment of speech sound disorders.

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